2	EST	40
3	Sessionals(May include Assignments/Projects/Tutorials/Quiz/ Lab evaluations)	35

ELECTIVE COURSE-3.2

URA625: ADVANCED HUMAN MACHINE INTERFACE				
	L	T	P	Cr
	2	2	0	3.0

Course Objective: This course is designed to provide you with a comprehensive understanding of Human-Machine Interfaces (HMIs). You will learn about the architecture and components of HMIs, develop skills in 3D game development using Unity 3D, create HMI applications using the QT framework, and gain expertise in Android and web app development. Additionally, you will explore the challenges and best practices for HMI testing and automation.

INTRODUCTION:

HMI Architecture and Concepts, HMI Subcomponents

GAMING ADVANCED 3D DEVELOPMENT:

Introduction to game development and advanced 3D development, Game Engine, Unity 3D installation -code editor - camera - game objects and transform, Renderer, lighting, UI, Scripting, Realtime 3D in the Automotive world, HMI Development.

QT

History of QT, Why QT? Supported Platforms, QT Installation, QT Creator, QT Modules, Signals and slots, Event Processing. HMI Application framework in QT, Creating and Testing of HMI application in QT.

ANDROID AND WEB APP DEVELOPMENT

Introduction to Android, Android application life cycle, Application development using Android framework and Testing.

HMI TESTING AND AUTOMATION

Introduction, elements of HMI, Challenges of HMI Testing, Verification and Validation

Course Learning Objectives (CLO)

The students will be able to:

1. summarize HMI architecture and its subcomponents.
2. develop real-time automotive applications using tools such as Unity and Qt.
3. develop simple HMI using android and web app development tools
4. perform HMI testing and validation for the developed system.

Text Books

1. Shuo Gao, Shuo Yan, Hang Zhao, Arokia Nathan, "Touch-Based Human-Machine Interaction: Principles and Applications", Springer Nature Switzerland AG; 1st Edition, 2021.
2. Robert Wells, "Unity 2020 By Example: A project-based guide to building 2D, 3D, augmented reality, and virtual reality games from scratch", Packt Publishing Limited, 2020.

Reference Books & Websites

1. Lee Zhi Eng, "Qt5 C++ GUI Programming Cookbook: Practical recipes for building cross-platform GUI applications, widgets, and animations with Qt 5, 2nd Edition, Packt Publishing Limited, 2019.
2. Karim Yaghmour, "Embedded Android: Porting, Extending, and Customizing", First Edition, Shroff/O'Reilly, 2013
3. Louis J.Williams, "Basic Programming Android for beginners Handbook", IT Campus Academy, March 2016
4. Julie C. Meloni, "Sams Teach Yourself HTML, CSS, and JavaScript All in One", Pearson Education, 2011.
5. Arnon Axelrod, "Complete Guide to Test Automation by Arnon Axelrod", Apress, September 2018.
6. Dean Alan Hume, "Progressive Web Apps", Manning Publisher, December 2017.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.